Assignment-1

What to Hand In

Use the provided assignment link.

Problem-3 (6 points):

Write a complete C++ program that prompts the user to enter an integer number and reads that number from the keyboard.

- If the entered number is negative, your program should print the following message: "The program doesn't accept negative numbers" and stop the execution.
- If the entered number is positive and less or equal 100 (between 0 to 100), the program should check and display whether the number is divisible by 4 or not. Also, your program **must** display quotient and remainder.
- If the entered number is greater than 100, your program should print square root of the given number.

  Here is a sample run:

  1)

```
Enter an integer number to see if the given number is divisible by 4 or not: -16
The program does not accept negative numbers
```

  2)

```
Please enter an integer number to see if it is divisible by 4: 16
The given number (16) is divisible by 4
Quotient: 4
Remainder: 0
```

  3)

```
Please enter an integer number to see if it is divisible by 4: 15
The given number (15) is not divisible by 4
Quotient: 3
Remainder: 3
```

4)

```
Enter an integer number to see if the given number is divisible by 4 or not: 121
The entered number is greater than 100 and square root of the number is 11
```

Problem-4 (6 points):

Write a complete C++ program that prompts the user to enter the side of a hexagon and displays its area. The area of a hexagon can be computed using the following formula (s is the length of a side):

$$Area = \frac{6 \times s^2}{4 \times \tan\left(\frac{\pi}{6}\right)}$$

Here is a sample run:

```
Enter the side: 5.5 ↵Enter
The area of the hexagon is 78.59
```

Use functions `fixed` and `setprecision` from **<iomanip>** header to set the precision of decimal numbers.

```
std::cout << std::fixed << std::setprecision(2) << 84.9384 <<
std::endl;


The output:
84.94
```

Use math functions from **<cmath>** header to get the value of Pi and tangent function.

Problem-5 (5 points):

Write a program that prompts the user to enter a string and displays the characters at odd index positions (1, 3, 5 …). Use **getline** function to read a string from an input stream.

Here is a sample run:

```
Enter a string: Programming is fun!
rgamn sfn
```

Problem-6 (7 points):

Write a program that reads an integer between 0 and 1000000 and adds all the digits in the integer. For example, if an integer is 932, the sum of all its digits is 14.

Hint: Use the % operator to extract digits and use the / operator to remove the extracted digit. For instance, 932 % 10 = 2 and 932 / 10 = 93.

Here is a sample run:

```
Enter an integer between 0 and 1000000: 123456
The sum of the digits is 21.
```

Bonus (+2 bonus points). Repeatedly add all digits of the entered number until the result has only one digit and display it.

Example:

Input: 38

Output: 2

Explanation: The process is

38 --> 3 + 8 --> 11

11 --> 1 + 1 --> 2

Since 2 has only one digit, display it.

```
Enter an integer between 0 and 1000000: 38
The sum of digits is 11.
The repeated sum of digits is 2.
```